

Quiz 11

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Let $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Solve the linear system $A\mathbf{x} = \mathbf{b}$ with

$\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ by Cramer's Rule.

$$\det A = \begin{vmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} = -1 \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 3 \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = (-1) \cdot (-1) + 3 \cdot (1-0) = 4$$

$$\det B_1 = \begin{vmatrix} 1 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = 1 \begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} = -1$$

$$\det B_2 = \begin{vmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{vmatrix} = (-1) \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1$$

$$\det B_3 = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{vmatrix} = 1 \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1$$

$$\Rightarrow x_1 = \frac{\det B_1}{\det A} = -\frac{1}{4}, \quad x_2 = \frac{\det B_2}{\det A} = \frac{1}{4}, \quad x_3 = \frac{\det B_3}{\det A} = \frac{1}{4}$$

Problem 2. (4 points) Use the determinant to compute the area of the triangle formed by the points $(1, 0)$, $(3, 3)$ and $(2, 4)$ in the plane.

$$\begin{aligned} \text{Area} &= \frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ 3 & 3 & 1 \\ 2 & 4 & 1 \end{vmatrix} = \frac{1}{2} \left[1 \begin{vmatrix} 3 & 1 \\ 4 & 1 \end{vmatrix} + 1 \begin{vmatrix} 3 & 3 \\ 2 & 4 \end{vmatrix} \right] \cdot \frac{1}{2} \\ &= \frac{1}{2} \left[1 \cdot (-1) + 1 \cdot (6) \right] \cdot \frac{1}{2} \\ &= \frac{5}{2} \end{aligned}$$